PortFlow Container Terminal: Yard and Gate Flow

PORTFLOW · Supply Chain and Logistics Engineering · Strategy Consulting Hub

Decision

Option B — Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot — delivers 21.7% lower year-5 yard utilization (target 19.5%) for CAD $9,400,000 of the CAD $10,000,000 budget: NPV CAD $6,091,191 at 9.5%, payback 32.7 months. Five-year net financial impact (case formula): -CAD $16,100,000 (net savings).

Why this option

Which package, within the $10M upfront budget, attacks PortFlow's core issue so that at +20% volume the yard runs at or below 85% utilization and trucks turn in under 45 minutes by year five?

Opt	Alternative	Capex	Gain	In budget	Meets target	NPV	Payback	Score /10
A	Static re-zoning only: one-off MIP and CRAFT layout	CAD $1,600,000	3.9%	yes	no	CAD $4,032,937	20.0 months	5.22
B ✔	Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot	CAD $9,400,000	21.7%	yes	yes	CAD $6,091,191	32.7 months	6.92
C	Yard expansion on the 15-acre parcel with two added RTGs	CAD $14,500,000	14.3%	NO	no	-CAD $14,627,035	> 60 months	2.9

Economics

Capex	NPV @ WACC	IRR	Payback	Benefit/ Cost	Run-rate benefit	Levelized cost
CAD $9,400,000	CAD $6,091,191	28.1%	32.7 months	1.42	CAD $6,400,000/ yr	134474.76 CAD per year per % point of improvement

Bear case (performance -15%, capex +20%): NPV CAD $140,445, payback 46.7 months.

Plan

- **Phase 1 (Months 1-3)** — Design: data, layout model, AGV safety case, CAD $2,200,000. KPI: Move data cleaned; MIP/CRAFT layout validated in simulation; AGV vendor and safety case approved
- **Phase 2 (Months 4-9)** — Re-zone, go dynamic, pilot 2 AGVs, CAD $5,600,000. KPI: Dwell <= 5.3 days; turn <= 55 min; rehandles -20%; AGVs commissioned
- **Phase 3 (Months 10-12)** — Scale decision and hand over, CAD $1,600,000. KPI: Dwell <= 4.8 days; turn < 45 min; utilization on track for <= 85%

Top judge questions

Q: Why not just build on the 15-acre parcel?
Space alone fails the objective: 25,344 / 28,000 TEU = 90.5% at year 5, for about $14.5M, over budget. Cutting dwell to 4.7 days reaches about 83% on today's footprint.

Q: Why only two AGVs?
The evidence is mixed: successful automated terminals cut operating costs 25-55%, but some lose 7-15% productivity. Two units in a fenced lane prove uptime, cost per move and safety for $1.6M; the fleet grows only if the month-12 gate passes.

Q: How do the MIP and CRAFT work together?
The MIP assigns container groups to blocks, minimising move frequency x distance to their exit within capacity and segregation rules. CRAFT then tries pairwise zone swaps and keeps any that lower total flow x distance, so the layout can be re-improved as demand shifts.